

score_forecasts

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Load in truth data and forecasts

We load in the hospitalization truth data and our forecasts. (Note: The load forecast parentheses function wasn't working, so alternate code is written to achieve the same output.)

```
full_hosp_truth <- load_truth("HealthData",
                              "inc hosp",
                              temporal_resolution="daily",
                              data_location = "remote_hub_repo")

inc_hosp_targets <- paste(0:35, "day ahead inc hosp")

load("../data/loadedfc_scores_df.RData")

# models <- c("Base_arima_4root", "Base_arima_noTrans",
#             "Multiple3_arima_4root", "Multiple3_arima_noTrans",
#             "Topmost8_arima_4root", "Topmost8_arima_noTrans")
#
# files <- rep("path", 30)
# for (i in 1:6) {
#   files <- list.files(path=paste("../data/", models[i], "/", sep=""), pattern=".csv", full.names=TRUE)
#   for (j in 1:30) {
#     if(i + j != 2) {
#       df <- read_csv(files[j]) %>%
#         mutate(model = models[i])
#       forecasts <- rbind(forecasts, df)
#     } else {
#       forecasts <- read_csv(files[j]) %>%
#         mutate(model = models[i])
#     }
#   }
# }
#
# forecasts_df <-
#   forecasts %>%
#     separate(target,
#              into=c("horizon", "temp"),
#              sep=" "
#             ) %>%
#     mutate(
```

```
#      horizon=as.numeric(horizon),
#      temporal_resolution="day",
#      target_variable="inc hosp"
#    ) %>%
#    select(model, forecast_date, location, horizon, temporal_resolution, target_variable, target_end)
#    left_join(hub_locations, by=c("location"="fips"))
#
# save(forecasts_df, file="../data/loadedfc_df.RData")
```

Plot Forecasts

We plot forecasts for three locations (Us National, Texas, and Vermont) for each of the three models. The two state locations were chosen as the highest and lowest hospitalization count locations. The plotted forecasts are spaced 4 weeks apart with 1 - 28 day ahead horizons, starting with the first forecast date, 12/07/20, until the end of the training period, 07/05/21.

```
locs <- full_hosp_truth %>%
  filter(target_end_date <= as.Date("2021-07-05")) %>%
  group_by(location) %>%
  summarize(cum_value=sum(value)) %>%
  ungroup() %>%
  arrange(desc(cum_value)) %>%
  filter(row_number() %in% c(1, 2, 53)) %>%
  pull(location)

fc_dates <- c(as.Date("2020-12-07") + weeks(4*(0:7)))

for (i in 1:3) {
  fdat <- forecasts_df %>% filter(location == locs[i], forecast_date %in% fc_dates)

  p <- plot_forecasts(fdat,
    target_variable = "inc hosp",
    intervals = c(.5, .95),
    truth_source = "HealthData",
    use_median_as_point = TRUE,
    facet = model ~.,
    facet_nrow = 3,
    facet_scales = "free_y",
    fill_by_model = TRUE,
    plot=FALSE)

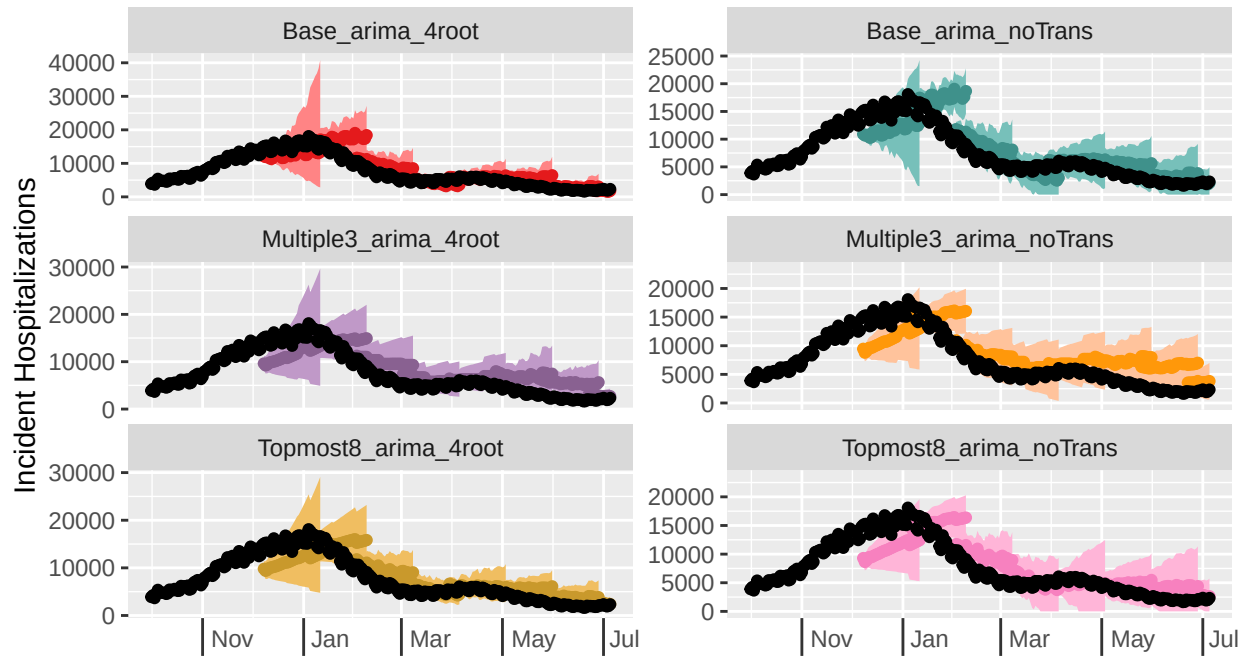
  pt <- p +
    scale_x_date(name=NULL, limits=c(as.Date("2020-10-01"), as.Date("2021-07-05")),
      date_breaks = "2 months", date_labels = "%b") +
    #scale_y_continuous(limits=c(0, max(filter(full_hosp_truth, location==locs[i])$value) * 1.25)) +
    theme(axis.ticks.length.x = unit(0.5, "cm"),
      axis.text.x = element_text(vjust = 7, hjust = -0.2),
      legend.position = "none")

  print(pt)
}
```

Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): United States

Selected forecast date(s): 2020-12-07, 2021-01-04, 2021-02-01, 2021-03-01, 2021-04-01, 2021-05-01, 2021-06-01, 2021-07-01

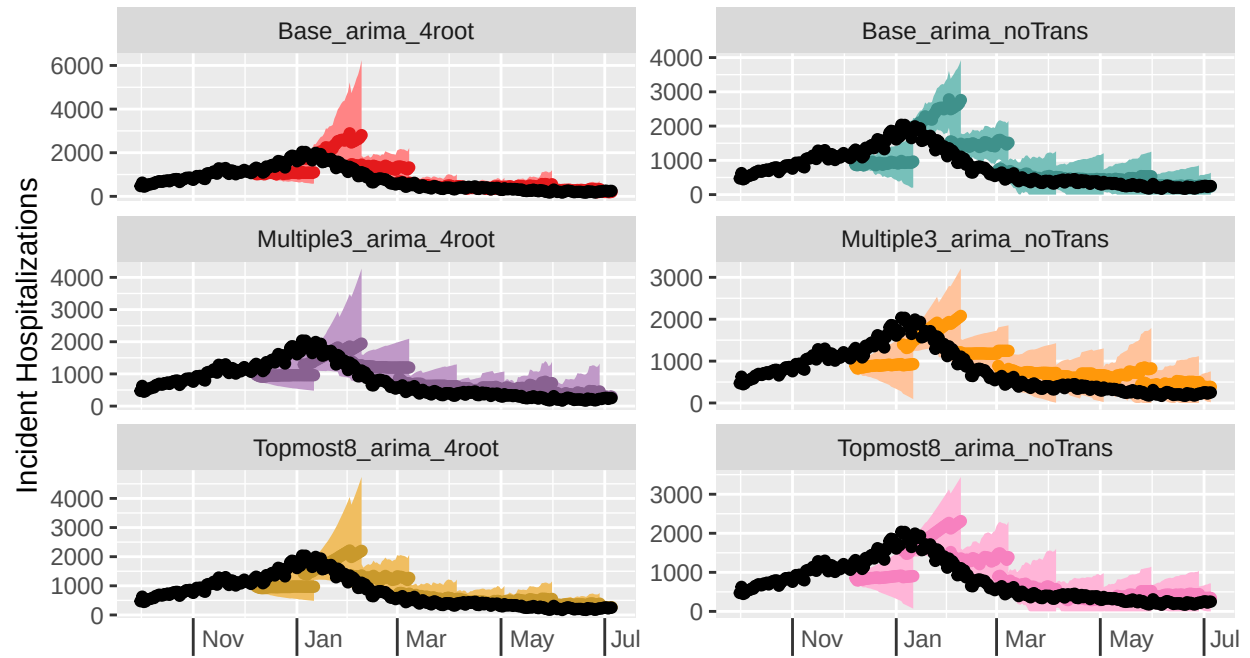


ans, Multiple3_arma_4root, Multiple3_arma_noTrans, Topmost8_arma_4root, Topmost8_arma_noTrans (forecasts)

Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): Texas

Selected forecast date(s): 2020-12-07, 2021-01-04, 2021-02-01, 2021-03-01, 2021

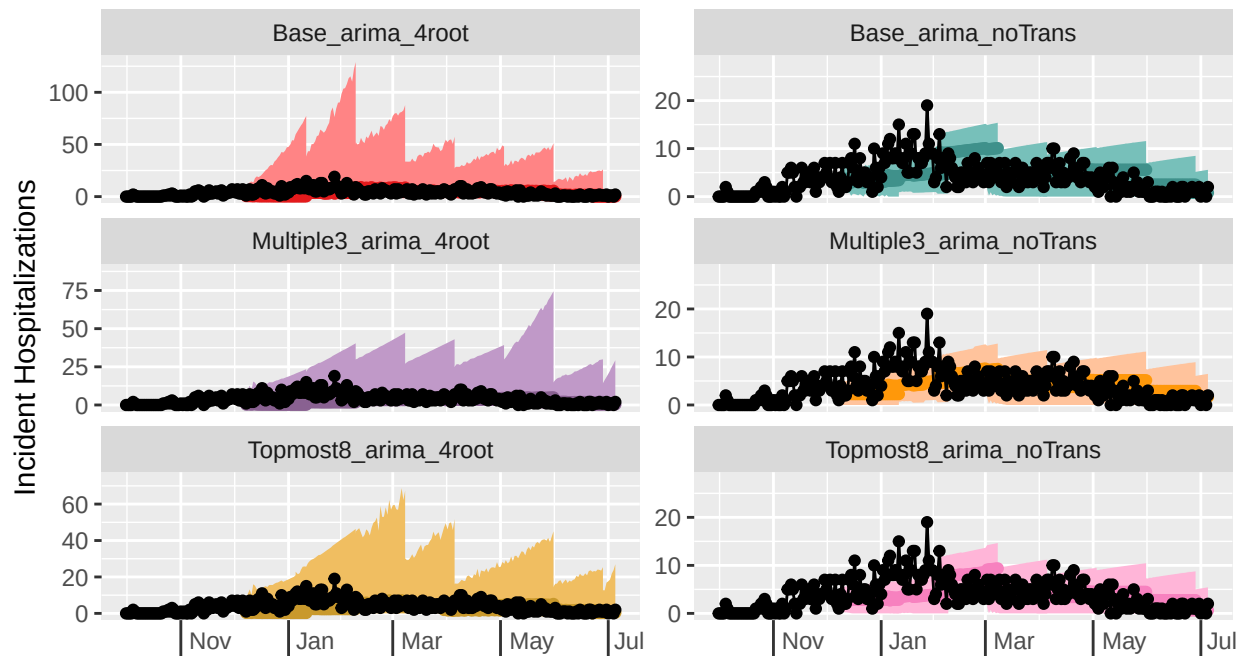


ans, Multiple3_arma_4root, Multiple3_arma_noTrans, Topmost8_arma_4root, Topmost8_arma_noTrans (forecasts)

Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): Vermont

Selected forecast date(s): 2020-12-07, 2021-01-04, 2021-02-01, 2021-03-01, 2021-



ans, Multiple3_arma_4root, Multiple3_arma_noTrans, Topmost8_arma_4root, Topmost8_arma_noTrans (forecasts)

The plots don't show much difference between the 6 models (although this is partially due to the fact that the y-scales are free and are not consistent even for the same location). However, of note is the fact that the models that use a forth root transformation tend to have wider prediction intervals than the models without transformations. All of the models reasonably follow the trend of the truth lines for every location shown, although it seems like lower count locations may have slightly greater accuracy. We investigate model accuracy more deeply by scoring the forecasts for each model and calculating appropriate metrics.

Score forecasts

We calculate metrics for all three models, aggregating by horizon. Since hospitalizations are forecasted at a daily time scale, a new horizon week variable is created in which scoring metrics are averaged over from a single week Tuesday to the following Monday. We perform these calculations for the entire training phase as well as for each of the two waves that occurred during the training phase.

```
scores_df <- score_forecasts(forecasts=forecasts_df, return_format="wide", truth=full_hosp_truth, use_m
#getwd()
#save(scores_df, file="../data/loadedfc_df.RData")
```

Overall metrics

```
# US National
overall_metrics_US <- scores_df %>%
  filter(horizon <= 28,
```

```

    target_end_date <= as.Date("2021-07-05"),
    location == "US") %>%
mutate(
  horizon_wk = case_when(
    horizon %in% c(1:7) ~ 1,
    horizon %in% c(8:14) ~ 2,
    horizon %in% c(15:21) ~ 3,
    horizon %in% c(22:28) ~ 4
  )
) %>%
group_by(model, horizon_wk) %>%
summarize(
  wis = mean(wis), mae=mean(abs_error),
  coverage_50 = mean(coverage_50),
  coverage_95 = mean(coverage_95)
)

```

'summarise()' has grouped output by 'model'. You can override using the '.groups' argument.

```
knitr::kable(overall_metrics_US)
```

model	horizon_wk	wis	mae	coverage_50	coverage_95
Base_arma_4root	1	639.3475	972.0104	0.2571429	0.7857143
Base_arma_4root	2	809.3146	1255.3891	0.4088670	0.8768473
Base_arma_4root	3	1156.8820	1765.0978	0.3367347	0.8265306
Base_arma_4root	4	1567.6289	2346.3781	0.2592593	0.7724868
Base_arma_noTrans	1	749.8177	1114.8615	0.2952381	0.8238095
Base_arma_noTrans	2	928.6026	1332.6191	0.4532020	0.8275862
Base_arma_noTrans	3	1200.9863	1718.6517	0.4897959	0.8469388
Base_arma_noTrans	4	1551.4460	2292.9731	0.3703704	0.8571429
Multiple3_arma_4root	1	1578.4370	2140.2091	0.0523810	0.3523810
Multiple3_arma_4root	2	1385.5491	2098.4276	0.1576355	0.6256158
Multiple3_arma_4root	3	1416.4651	2197.3134	0.2244898	0.7091837
Multiple3_arma_4root	4	1763.8126	2694.8060	0.2328042	0.5820106
Multiple3_arma_noTrans	1	1618.6213	2246.3867	0.0952381	0.4714286
Multiple3_arma_noTrans	2	1438.8029	2266.6343	0.1083744	0.7389163
Multiple3_arma_noTrans	3	1482.0367	2403.1285	0.2193878	0.8520408
Multiple3_arma_noTrans	4	1906.5512	3012.1649	0.2010582	0.7619048
Topmost8_arma_4root	1	1088.9810	1550.9345	0.1142857	0.4666667
Topmost8_arma_4root	2	1023.5207	1616.6941	0.2167488	0.7783251
Topmost8_arma_4root	3	1212.1641	1919.1710	0.2500000	0.7551020
Topmost8_arma_4root	4	1470.9185	2299.8693	0.2539683	0.6825397
Topmost8_arma_noTrans	1	1044.4676	1506.7465	0.2190476	0.7333333
Topmost8_arma_noTrans	2	1116.5918	1740.3537	0.3054187	0.8177340
Topmost8_arma_noTrans	3	1298.1247	2080.3672	0.3214286	0.8469388
Topmost8_arma_noTrans	4	1588.5448	2432.3564	0.3809524	0.8306878

```

# Aggregated for all states and territories
overall_metrics_states <- scores_df %>%
  filter(horizon <= 28,
    target_end_date <= as.Date("2021-07-05"),

```

```

    location != "US") %>%
mutate(
  horizon_wk = case_when(
    horizon %in% c(1:7) ~ 1,
    horizon %in% c(8:14) ~ 2,
    horizon %in% c(15:21) ~ 3,
    horizon %in% c(22:28) ~ 4
  )
) %>%
group_by(model, horizon_wk) %>%
summarize(
  wis = mean(wis), mae=mean(abs_error),
  coverage_50 = mean(coverage_50),
  coverage_95 = mean(coverage_95)
)

```

‘summarise()’ has grouped output by ‘model’. You can override using the ‘.groups’ argument.

```
knitr::kable(overall_metrics_states)
```

model	horizon_wk	wis	mae	coverage_50	coverage_95
Base_arma_4root	1	19.20331	28.51321	0.4781136	0.9009158
Base_arma_4root	2	23.30803	35.77387	0.5017052	0.9324555
Base_arma_4root	3	31.05692	47.77089	0.4797881	0.9233713
Base_arma_4root	4	39.58292	60.27320	0.4624542	0.9059829
Base_arma_noTrans	1	21.44460	31.40634	0.5219780	0.9112637
Base_arma_noTrans	2	25.36247	38.11666	0.5804282	0.9277188
Base_arma_noTrans	3	32.44306	48.42309	0.5897763	0.9182692
Base_arma_noTrans	4	40.14937	59.56382	0.5810948	0.9055759
Multiple3_arma_4root	1	34.01662	46.97430	0.3177656	0.7446886
Multiple3_arma_4root	2	34.21900	48.86137	0.3691739	0.8471012
Multiple3_arma_4root	3	45.00629	52.95103	0.3924647	0.8843210
Multiple3_arma_4root	4	90.81650	63.98877	0.3803419	0.8734229
Multiple3_arma_noTrans	1	33.78419	48.46517	0.3260073	0.8269231
Multiple3_arma_noTrans	2	33.50921	52.08516	0.3800682	0.9217507
Multiple3_arma_noTrans	3	36.46624	58.69374	0.4084576	0.9398548
Multiple3_arma_noTrans	4	44.74088	72.38738	0.3937729	0.9313187
Topmost8_arma_4root	1	26.03848	37.30614	0.3787546	0.8128205
Topmost8_arma_4root	2	26.43012	40.32504	0.4272452	0.8986358
Topmost8_arma_4root	3	30.69040	47.89080	0.4245487	0.9144427
Topmost8_arma_4root	4	36.82166	57.17876	0.4252137	0.9078144
Topmost8_arma_noTrans	1	25.96863	36.81447	0.4780220	0.8846154
Topmost8_arma_noTrans	2	27.59916	40.95389	0.5346722	0.9244979
Topmost8_arma_noTrans	3	32.35673	49.56247	0.5323783	0.9276884
Topmost8_arma_noTrans	4	37.99783	58.30469	0.5484330	0.9189052

When splitting the results so that US National is separate, we find that for US National the two models that use the base set hierarchy tended to perform the best, followed by the Topmost8_arma_noTrans model. The Base_arma_noTrans model performed the best in terms of PI Coverage, followed by the Base_arma_4root and Topmost8_arma_noTrans models (with the latter being more consistent with coverage rates for all horizons). In terms of WIS and MAE, the Base_arma_4root model performed the best, followed by the

Base_arma_noTrans model. Overall, for the US National level, it seems that there is a pattern that for coverage rates, the models without transformations perform better (and these tend to have smaller prediction intervals) while for WIS and MAE the fourth root transform models perform better; these patterns hold at all horizons. Further, there is a general order to model accuracy in terms of hierarchical structure used: the Base set performs the best, followed by the Topmost8 set, and finally the Multiple3 set.

For the state level, we see noticeably better coverage rates that approach the nominal level (or even exceed it) for all of the models. (We also see shrunken WIS and MAE values, although this is likely partially due to the fact that the states have smaller scale values than the National level.) It's a little harder to definitively say which models have the best coverage rates, as the results differ between the 50% and 95% intervals and the models are quite similar in coverage rates, especially for the 95% intervals. However, it seems that the Base_arma_4root model has the best performance for all metrics across all horizons. It is followed closely by the Topmost8_arma_noTrans and Base_arma_noTrans models, then the Topmost8_arma_4root and Multiple3_arma_noTrans models, all of which are much better than the Multiple3_arma_4root model.

Metrics by wave

```
# US National
wave_metrics_US <- scores_df %>%
  filter(horizon <= 28,
         target_end_date <= as.Date("2021-07-05"),
         location == "US") %>%
  mutate(
    horizon_wk = case_when(
      horizon %in% c(1:7) ~ 1,
      horizon %in% c(8:14) ~ 2,
      horizon %in% c(15:21) ~ 3,
      horizon %in% c(22:28) ~ 4
    ),
    wave = case_when(
      forecast_date <= as.Date("2021-03-17") ~ "winter21",
      forecast_date >= as.Date("2021-03-17") ~ "alpha"
    )
  ) %>%
  group_by(wave, model, horizon_wk) %>%
  summarize(
    wis = mean(wis), mae=mean(abs_error),
    coverage_50 = mean(coverage_50),
    coverage_95 = mean(coverage_95)
  )
```

'summarise()' has grouped output by 'wave', 'model'. You can override using the '.groups' argument.

```
knitr::kable(wave_metrics_US)
```

wave	model	horizon_wk	wis	mae	coverage_50	coverage_95
alpha	Base_arma_4root	1	348.2289	543.0210	0.2952381	0.8190476
alpha	Base_arma_4root	2	450.9277	703.5284	0.4285714	0.9387755
alpha	Base_arma_4root	3	647.7496	1044.3615	0.3186813	0.9010989
alpha	Base_arma_4root	4	827.5967	1349.3859	0.2261905	0.8571429

wave	model	horizon_wk	wis	mae	coverage_50	coverage_95
alpha	Base_arima_noTrans	1	403.4274	678.4257	0.4190476	1.0000000
alpha	Base_arima_noTrans	2	534.5930	827.9246	0.5102041	1.0000000
alpha	Base_arima_noTrans	3	689.8029	1088.3635	0.5494505	1.0000000
alpha	Base_arima_noTrans	4	897.8446	1508.1117	0.4404762	1.0000000
alpha	Multiple3_arima_4root	1	1333.0922	1769.3234	0.0285714	0.3238095
alpha	Multiple3_arima_4root	2	1209.9805	1749.9669	0.1734694	0.5510204
alpha	Multiple3_arima_4root	3	1391.1703	2040.6824	0.1538462	0.4945055
alpha	Multiple3_arima_4root	4	1782.3330	2639.0030	0.1190476	0.3571429
alpha	Multiple3_arima_noTrans	1	1526.8979	2190.9790	0.0857143	0.4476190
alpha	Multiple3_arima_noTrans	2	1456.7269	2357.6243	0.0816327	0.7040816
alpha	Multiple3_arima_noTrans	3	1692.6105	2754.0069	0.1538462	0.8901099
alpha	Multiple3_arima_noTrans	4	2187.8663	3535.4044	0.0595238	0.7380952
alpha	Topmost8_arima_4root	1	637.3661	943.9450	0.1142857	0.4666667
alpha	Topmost8_arima_4root	2	653.8217	1010.8866	0.2653061	0.7346939
alpha	Topmost8_arima_4root	3	887.1828	1363.9112	0.2197802	0.6703297
alpha	Topmost8_arima_4root	4	1072.7247	1672.5638	0.1904762	0.6190476
alpha	Topmost8_arima_noTrans	1	546.7007	952.7876	0.2380952	0.9809524
alpha	Topmost8_arima_noTrans	2	645.8991	1124.3019	0.4387755	1.0000000
alpha	Topmost8_arima_noTrans	3	848.2689	1468.6578	0.4835165	1.0000000
alpha	Topmost8_arima_noTrans	4	1011.0640	1649.8331	0.4761905	1.0000000
winter21	Base_arima_4root	1	930.4661	1400.9997	0.2190476	0.7523810
winter21	Base_arima_4root	2	1143.8090	1770.4590	0.3904762	0.8190476
winter21	Base_arima_4root	3	1598.1301	2389.7358	0.3523810	0.7619048
winter21	Base_arima_4root	4	2159.6547	3143.9718	0.2857143	0.7047619
winter21	Base_arima_noTrans	1	1096.2081	1551.2974	0.1714286	0.6476190
winter21	Base_arima_noTrans	2	1296.3448	1803.6672	0.4000000	0.6666667
winter21	Base_arima_noTrans	3	1644.0119	2264.9014	0.4380952	0.7142857
winter21	Base_arima_noTrans	4	2074.3271	2920.8622	0.3142857	0.7428571
winter21	Multiple3_arima_4root	1	1823.7818	2511.0948	0.0761905	0.3809524
winter21	Multiple3_arima_4root	2	1549.4132	2423.6576	0.1428571	0.6952381
winter21	Multiple3_arima_4root	3	1438.3873	2333.0603	0.2857143	0.8952381
winter21	Multiple3_arima_4root	4	1748.9964	2739.4484	0.3238095	0.7619048
winter21	Multiple3_arima_noTrans	1	1710.3447	2301.7945	0.1047619	0.4952381
winter21	Multiple3_arima_noTrans	2	1422.0739	2181.7103	0.1333333	0.7714286
winter21	Multiple3_arima_noTrans	3	1299.5395	2099.0338	0.2761905	0.8190476
winter21	Multiple3_arima_noTrans	4	1681.4991	2593.5733	0.3142857	0.7809524
winter21	Topmost8_arima_4root	1	1540.5959	2157.9240	0.1142857	0.4666667
winter21	Topmost8_arima_4root	2	1368.5731	2182.1146	0.1714286	0.8190476
winter21	Topmost8_arima_4root	3	1493.8145	2400.3962	0.2761905	0.8285714
winter21	Topmost8_arima_4root	4	1789.4736	2801.7137	0.3047619	0.7333333
winter21	Topmost8_arima_noTrans	1	1542.2344	2060.7054	0.2000000	0.4857143
winter21	Topmost8_arima_noTrans	2	1555.9051	2315.3353	0.1809524	0.6476190
winter21	Topmost8_arima_noTrans	3	1687.9997	2610.5152	0.1809524	0.7142857
winter21	Topmost8_arima_noTrans	4	2050.5295	3058.3751	0.3047619	0.6952381

```

# Aggregated for all states and territories
wave_metrics_states <- scores_df %>%
  filter(horizon <= 28,
         target_end_date <= as.Date("2021-07-05"),
         location != "US") %>%
  mutate(
    horizon_wk = case_when(

```

```

    horizon %in% c(1:7) ~ 1,
    horizon %in% c(8:14) ~ 2,
    horizon %in% c(15:21) ~ 3,
    horizon %in% c(22:28) ~ 4
  ),
  wave = case_when(
    forecast_date <= as.Date("2021-03-17") ~ "winter21",
    forecast_date >= as.Date("2021-03-17") ~ "alpha"
  )
) %>%
group_by(wave, model, horizon_wk) %>%
summarize(
  wis = mean(wis), mae=mean(abs_error),
  coverage_50 = mean(coverage_50),
  coverage_95 = mean(coverage_95)
)

```

'summarise()' has grouped output by 'wave', 'model'. You can override using the '.groups' argument.

```
knitr::kable(wave_metrics_states)
```

wave	model	horizon_wk	wis	mae	coverage_50	coverage_95
alpha	Base_arima_4root	1	9.401447	14.53343	0.4968864	0.9159341
alpha	Base_arima_4root	2	12.182470	18.80838	0.5198195	0.9503532
alpha	Base_arima_4root	3	16.393942	25.75905	0.5021133	0.9547760
alpha	Base_arima_4root	4	20.932534	32.67461	0.4896978	0.9500916
alpha	Base_arima_noTrans	1	11.704930	18.22464	0.6565934	0.9895604
alpha	Base_arima_noTrans	2	15.171337	23.44738	0.7115385	0.9927394
alpha	Base_arima_noTrans	3	19.082068	29.60696	0.7278107	0.9900676
alpha	Base_arima_noTrans	4	23.544684	36.95682	0.7152015	0.9890110
alpha	Multiple3_arima_4root	1	20.895309	30.36914	0.3111722	0.7351648
alpha	Multiple3_arima_4root	2	25.404048	34.26344	0.3563579	0.8392857
alpha	Multiple3_arima_4root	3	49.093669	40.97925	0.3609467	0.8683432
alpha	Multiple3_arima_4root	4	145.357628	53.06648	0.3326465	0.8566850
alpha	Multiple3_arima_noTrans	1	26.512088	41.94125	0.2968864	0.8582418
alpha	Multiple3_arima_noTrans	2	28.285180	47.97319	0.3304553	0.9640895
alpha	Multiple3_arima_noTrans	3	32.777879	56.29881	0.3560862	0.9771767
alpha	Multiple3_arima_noTrans	4	41.064825	70.59136	0.3331044	0.9771062
alpha	Topmost8_arima_4root	1	13.596337	20.84621	0.3816850	0.8186813
alpha	Topmost8_arima_4root	2	15.042247	23.94270	0.4228807	0.9054160
alpha	Topmost8_arima_4root	3	19.133755	30.76062	0.4110313	0.9285714
alpha	Topmost8_arima_4root	4	23.110194	37.05173	0.4068223	0.9358974
alpha	Topmost8_arima_noTrans	1	13.384423	20.87681	0.6080586	0.9738095
alpha	Topmost8_arima_noTrans	2	15.851603	23.42293	0.7023155	0.9876374
alpha	Topmost8_arima_noTrans	3	19.690620	29.79342	0.7035080	0.9917582
alpha	Topmost8_arima_noTrans	4	23.026581	35.65114	0.7163462	0.9956502
winter21	Base_arima_4root	1	29.005164	42.49299	0.4593407	0.8858974
winter21	Base_arima_4root	2	33.691894	51.60833	0.4847985	0.9157509
winter21	Base_arima_4root	3	43.764830	66.84781	0.4604396	0.8961538
winter21	Base_arima_4root	4	54.503222	82.35207	0.4406593	0.8706960
winter21	Base_arima_noTrans	1	31.184266	44.58804	0.3873626	0.8329670

wave	model	horizon_wk	wis	mae	coverage_50	coverage_95
winter21	Base_arma_noTrans	2	34.874197	51.80799	0.4580586	0.8670330
winter21	Base_arma_noTrans	3	44.022591	64.73039	0.4701465	0.8560440
winter21	Base_arma_noTrans	4	53.433122	77.64941	0.4738095	0.8388278
winter21	Multiple3_arma_4root	1	47.137941	63.57946	0.3243590	0.7542125
winter21	Multiple3_arma_4root	2	42.446283	62.48610	0.3811355	0.8543956
winter21	Multiple3_arma_4root	3	41.463889	63.32657	0.4197802	0.8981685
winter21	Multiple3_arma_4root	4	47.183602	72.72661	0.4184982	0.8868132
winter21	Multiple3_arma_noTrans	1	41.056291	54.98909	0.3551282	0.7956044
winter21	Multiple3_arma_noTrans	2	38.384963	55.92299	0.4263736	0.8822344
winter21	Multiple3_arma_noTrans	3	39.662812	60.76934	0.4538462	0.9075092
winter21	Multiple3_arma_noTrans	4	47.681726	73.82420	0.4423077	0.8946886
winter21	Topmost8_arma_4root	1	38.480624	53.76607	0.3758242	0.8069597
winter21	Topmost8_arma_4root	2	37.058800	55.61523	0.4313187	0.8923077
winter21	Topmost8_arma_4root	3	40.706153	62.73696	0.4362637	0.9021978
winter21	Topmost8_arma_4root	4	47.790834	73.28038	0.4399267	0.8853480
winter21	Topmost8_arma_noTrans	1	38.552836	52.75213	0.3479853	0.7954212
winter21	Topmost8_arma_noTrans	2	38.563538	57.31612	0.3782051	0.8655678
winter21	Topmost8_arma_noTrans	3	43.334026	66.69565	0.3840659	0.8721612
winter21	Topmost8_arma_noTrans	4	49.974836	76.42753	0.4141026	0.8575092

For US National during the alpha wave, the Base_arma_noTrans model performed the best for coverage rates and second best for WIS and MAE at every horizon. The Base_arma_4root model performed best in terms of WIS and MAE but was third for coverage rate while the Topmost8_arma_noTrans model had the second best coverage rates and third best WIS and MAE. During the alpha wave, these models performed the best and tended to have more consistent values for their metrics. The winter21 wave saw noticeably worse values for all metrics at every horizon. Generally, the two models that use the base set hierarchy tended to perform the best for all metrics, although the coverage rates varied a lot by horizon, with the Topmost8_arma_4root and Multiple3_arma_noTrans achieving closer to nominal coverage rates for the 95% interval than even the Base models at horizons greater than 1 week ahead.

For states during both waves, we again see noticeably better coverage rates that approach the nominal level (or even exceed it) for all of the models, compared to those of the US National level when broken into waves. During the alpha wave, the Base_arma_4root model has the best performance for all metrics across all horizons. It is followed closely by the Topmost8_arma_noTrans and Base_arma_noTrans models, then the Topmost8_arma_4root and Multiple3_arma_noTrans models, all of which are much better than the Multiple3_arma_4root model. During the winter21 wave, the Base_arma_4root model is again generally the best for all metrics at all horizon weeks but sometimes loses out in terms of 95% coverage at longer horizons to other models. Notably during this wave, there are much greater similarities in coverage rates and certain models have better coverage rates for the 50% interval while others have better coverage reads for the 95% one.

Scoring Conclusions

It seems that the best three models that should be passed onto the testing phase are the Base_arma_noTrans, the Base_arma_4root, and the Topmost8_arma_noTrans models, which seem to take turns being most accurate depending on state or national level overall and during different waves.